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MTH:2310 DISCRETE MATH
QUIZ 7
DUE: MONDAY APRIL 7

Complete all of the following problems being as detailed as you can be. You may work in groups, but you must submit your own solutions using your own words. You must submit a physical copy of your solutions to me unless told otherwise. If there is a specific number of problems which seem unsolvable let me know first, I may have made a typo. Even if you cannot completely solve a problem, a good description of key terms and concepts relevant to the problem is sufficient for some partial credit.

Handwriting: *(4 points)*
Your ability to express your solutions is just as important as mathematical accuracy.
You will be penalized up to 2 points for arithmetic errors.
You will be penalized up to 2 points for disorganized or hard to read solutions.

Problem 1. *(16 points) Solving Linear Recurrence Relations*
Suppose $(r+2)^2(2r-3)^3(r-1)(r+1)(r-7)^5 = 0$ is the characteristic equation for a recurrence relation.

(a) *(2 points)* What is the degree of the recurrence relation?

$2+3+1+1+5 = \boxed{12}$

(b) *(4 points)* Find all characteristic roots and state their multiplicity.

$r_1 = 2 \quad m_1 = 2 \qquad r_4 = -1 \quad m_4 = 1$

$r_2 = \frac{3}{2} \quad m_2 = 3 \qquad r_5 = 7 \quad m_5 = 5$

$r_3 = 1 \quad m_3 = 1$

(c) *(10 points)* What is the general form of the solution to the recurrence relation?

$$\Big\{\alpha_1(2)^n + \alpha_2 n(2)^n + \alpha_3\left(\tfrac{3}{2}\right)^n + \alpha_4 n\left(\tfrac{3}{2}\right)^n + \alpha_5 n^2\left(\tfrac{3}{2}\right)^n + \alpha_6(1)^n + \alpha_7(-1)^n + \alpha_8(7)^n + \alpha_9 n(7)^n + \alpha_{10} n^2(7)^n + \alpha_{11} n^3(7)^n + \alpha_{12} n^4(7)^n\Big\}_{n=0}^{\infty}$$

Problem 2. *(16 points) Solving Linear Recurrence Relations*
Find the solution to the recurrence relation $a_n = \frac{3}{2}a_{n-1} + 9a_{n-2} - \frac{27}{2}a_{n-3}$ with initial conditions $a_0 = 3$, $a_1 = 15$, $a_2 = 0$. Make sure to carefully organize your solution.

$$r^3 = \frac{3}{2}r^2 + 9r - \frac{27}{2} \Rightarrow 2r^3 - 3r^2 - 18r + 27 = 0$$

$$\Rightarrow r^2(2r-3) - 9(2r-3) = 0$$

$$\Rightarrow (r^2-9)(2r-3) = 0$$

So $r = \pm 3, \frac{3}{2}$

<u>General Form</u>: $\{\alpha_1(3)^n + \alpha_2(-3)^n + \alpha_3(\frac{3}{2})^n\}_{n=0}^{\infty}$

or $a_n = \alpha_1(3)^n + \alpha_2(-3)^n + \alpha_3(\frac{3}{2})^n \quad n \geq 0$

$$\begin{cases} 3 = \alpha_1 + \alpha_2 + \alpha_3 \\ 15 = 3\alpha_1 - 3\alpha_2 + \frac{3}{2}\alpha_3 \\ 0 = 9\alpha_1 + 9\alpha_2 + \frac{9}{4}\alpha_3 \end{cases} \xrightarrow[R_3]{-9R_1+R_3} \begin{cases} 3 = \alpha_1 + \alpha_2 + \alpha_3 \\ 15 = 3\alpha_1 - 3\alpha_2 + \frac{3}{2}\alpha_3 \\ -27 = 0 + 0 - \frac{27}{4}\alpha_3 \end{cases}$$

$$\xrightarrow[R_2]{-3R_1+R_2} \begin{cases} 3 = \alpha_1 + \alpha_2 + \alpha_3 \\ 6 = 0 - 6\alpha_2 - \frac{3}{2}\alpha_3 \\ 1 = 0 + 0 + \frac{1}{4}\alpha_3 \end{cases} \longrightarrow \boxed{\alpha_3 = 4}$$

So $6 = -6\alpha_2 - \frac{3}{2}(4) = -6\alpha_2 - 6 \Rightarrow 12 = -6\alpha_2 \Rightarrow \boxed{\alpha_2 = -2}$

So $3 = \alpha_1 - 2 + 4 = \alpha_1 + 2$ so $\boxed{\alpha_1 = 1}$

$$\{(3)^n - 2(-3)^n + 4(\tfrac{3}{2})^n\}_{n=0}^{\infty}$$

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Problem 3. *(16 points) Solving Linear Recurrence Relations*
Find the solution to the recurrence relation $a_n = 6a_{n-1} - 12a_{n-2} + 8a_{n-3}$ with initial conditions $a_0 = -2$, $a_1 = 2$, $a_2 = 24$. Make sure to carefully organize your solution.

$r^3 = 6r^2 - 12r + 8 \Rightarrow r^3 - 6r^2 + 12r - 8 = 0$

$$\begin{array}{r|rrrr} 2 & 1 & -6 & 12 & -8 \\ & & 2 & -8 & 8 \\ \hline & 1 & -4 & 4 & 0 \end{array} \Rightarrow (r-2)(r^2-4r+4) = 0$$

$\Rightarrow (r-2)(r-2)^2 = 0$

$r = 2$ with multiplicity 3.

General Form: $\{\alpha_1 (2)^n + \alpha_2 n(2)^n + \alpha_3 n^2 (2)^n\}_{n=0}^{\infty}$

or $a_n = \alpha_1 (2)^n + \alpha_2 n (2)^n + \alpha_3 n^2 (2)^n \quad n \geq 0$

$$\begin{cases} -2 = \alpha_1 \\ 2 = 2\alpha_1 + 2\alpha_2 + 2\alpha_3 \\ 24 = 4\alpha_1 + 8\alpha_2 + 16\alpha_3 \end{cases} \Rightarrow \begin{cases} -2 = \alpha_1 \\ 2 = -4 + 2\alpha_2 + 2\alpha_3 \\ 24 = -8 + 8\alpha_2 + 16\alpha_3 \end{cases}$$

$$\Rightarrow \begin{cases} -2 = \alpha_1 \\ 6 = 2\alpha_2 + 2\alpha_3 \\ 32 = 8\alpha_2 + 16\alpha_3 \end{cases} \xrightarrow[R_3]{-4R_2 + R_3} \begin{cases} -2 = \alpha_1 \\ 6 = 2\alpha_2 + 2\alpha_3 \\ 8 = + 8\alpha_3 \end{cases}$$

$8 = 8\alpha_3 \Rightarrow \boxed{\alpha_3 = 1}$

$6 = 2\alpha_2 + 2 \Rightarrow 4 = 2\alpha_2 \Rightarrow \boxed{2 = \alpha_2}$

$\{-2(2)^n + 2n(2)^n + n^2(2)^n\}_{n=0}^{\infty}$

Problem 4. *(14 points) Mathematical Induction*

Prove that for every positive integer n, $1^3 + 2^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2}\right)^2$. You will be graded on format as well as accuracy.

Let $P(n)$: " $1^3+2^3+\cdots+n^3 = \left(\frac{n(n+1)}{2}\right)^2$ " where n is a positive integer.

Base Case $(n=1)$: Consider $P(1)$: " $1^3 = \left(\frac{1\cdot(1+1)}{2}\right)^2$ "
where $\left(\frac{1\cdot(1+1)}{2}\right)^2 = \left(\frac{2}{2}\right)^2 = 1^2 = 1$ and $1^3 = 1$ so $P(1)$ is true.

Inductive Step: Assume $P(k)$ is true for an arbitrary positive integer k. $P(k)$: " $1^3+2^3+\cdots+k^3 = \left(\frac{k(k+1)}{2}\right)^2$ ".

We want to show $P(k+1)$ is true.
$P(k+1)$: " $1^3+2^3+\cdots+k^3+(k+1)^3 = \left(\frac{(k+1)((k+1)+1)}{2}\right)^2$. "

Notice
$$1^3+2^3+\cdots+k^3+(k+1)^3 = \left(\frac{k(k+1)}{2}\right)^2 + (k+1)^3 \text{ by } P(k).$$

$$\left(\frac{k(k+1)}{2}\right)^2 + (k+1)^3 = \frac{k^2(k+1)^2}{4} + (k+1)^3 = (k+1)^2\left(\frac{k^2}{4} + (k+1)\right)$$
$$= (k+1)^2\left(\frac{k^2+4k+4}{4}\right) = (k+1)^2\left(\frac{(k+2)^2}{4}\right) = \left(\frac{(k+1)(k+2)}{2}\right)^2$$
$$= \left(\frac{(k+1)((k+1)+1)}{2}\right)^2$$

So $P(k+1)$ is true.

Therefore, by mathematical induction, if n is any positive integer then $1^3+2^3+\cdots+n^3 = \left(\frac{n(n+1)}{2}\right)^2$ $\blacksquare$

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Problem 5. *(14 points) Mathematical Induction*
Prove that for any non-negative integer n, 24 divides $5^{2n} - 1$. You will be graded on format as well as accuracy.

Let $P(n)$: " $24 \mid 5^{2n}-1$ " where n is a non-negative integer.

Base case ($n=0$): Consider $P(0)$: " $24 \mid 5^{2(0)}-1$ " where $5^{2(0)}-1 = 1-1 = 0$ and $\frac{0}{24} = 0$ so $P(0)$ is true.

Inductive Step: Assume $P(k)$ is true for an arbitrary non-negative integer k. $P(k)$: " $24 \mid 5^{2k}-1$. "
We want to show $P(k+1)$ is true. $P(k+1)$: " $24 \mid 5^{2(k+1)}-1$. "

Notice

$$5^{2(k+1)}-1 = 5^{2k+2}-1 = 25(5^{2k})-1.$$

By $P(k)$, there is an integer c such that $5^{2k}-1 = 24c$ or $5^{2k} = 24c+1$.

Then

$$25(5^{2k})-1 = 25(24c+1)-1 = (25)(24)c + 25 - 1$$
$$= (25)(24)c + 24 = 24(25c+1)$$

so $5^{2(k+2)}-1$ is divisible by 24. Then $P(k+1)$ is true.

Therefore, by mathematical induction, if n is any non-negative integer, 24 divides $5^{2n}-1$ ∎